

From Genome to Drug Target: A Stepwise Computational Strategy for *Acinetobacter baumannii*

The aim of this project is to identify potential drug targets in *Acinetobacter baumannii* through a stepwise genome-wide computational approach. The complete genome GCF_009035845.1 was analyzed using Pyrodigal, which predicted 3,801 proteins that were used as the starting protein set. These proteins were compared with the Database of Essential Genes (DEG) using BLASTp, applying identity $\geq 30\%$, query coverage $\geq 70\%$, and E-value $\leq 1e-5$. This reduced the initial 3,801 proteins to 1,607 essentiality candidates.

The 1,607 proteins were then compared against protein sequences from 10 additional *A. baumannii* strains to evaluate conservation. Based on the same sequence-similarity criteria, 1,471 proteins were found to be conserved across all 10 strains and were retained as the core conserved protein set.

The 1,471 core conserved proteins were subsequently compared against the human proteome to identify and remove proteins with significant similarity to human proteins. This resulted in the removal of 379 potential human homologs, leaving 1,092 bacteria-specific proteins. These proteins were then functionally characterized using BlastKOALA and KEGG. A total of 881 proteins received KO annotations, representing 823 unique KO IDs. KEGG pathway mapping was used to connect these proteins with their biological functions and pathways.

Based on the resulting functional information, proteins associated with biologically important antibacterial processes, including cell wall/envelope, DNA replication and repair, protein synthesis, nucleotide metabolism, amino acid biosynthesis, and cofactor/vitamin metabolism, were prioritized. Through this functional and biological filtering, the candidate pool was further reduced to 295 proteins considered more relevant to antibacterial target discovery.

The 295 candidates were then subjected to additional target prioritization. Proteins representing redundant KO functions were examined to avoid selecting multiple proteins representing the same functional target. Candidates were further evaluated according to their antibacterial mechanisms and pathway relevance, with particular emphasis on cell wall/envelope, DNA replication/repair, and protein synthesis/ribosome-related targets. The candidates were ranked and grouped according to these mechanisms, resulting in a final shortlist of 15 preliminary target proteins.

These 15 proteins are now the focus of the next stage, where each candidate will be evaluated using literature evidence, biological function, previous inhibitor or drug studies, PDB/structural availability, binding-site characteristics, and overall druggability. This final validation will be used to narrow the 15 candidates to approximately 1–3 high-confidence drug targets for subsequent structural analysis, molecular docking, and virtual screening.